



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	The organization experienced a denial-of-service (DoS) attack that disrupted its internal network for approximately two hours. During the incident, a large flood of ICMP packets overwhelmed the network, causing network services to become unresponsive and preventing normal internal traffic from accessing network resources. The incident response team mitigated the attack by blocking incoming ICMP packets, taking non-critical network services offline, and restoring critical services to resume business operations. A subsequent investigation determined that a malicious actor exploited an unconfigured firewall to send excessive ICMP ping traffic into the network, successfully launching the DoS attack. To reduce the risk of similar incidents in the future, the organization implemented several security improvements, including firewall rules to limit ICMP traffic, source IP verification to detect spoofed addresses, network monitoring software to identify abnormal traffic patterns, and an IDS/IPS solution to filter suspicious ICMP traffic.
Identify	• The security team identified that an unconfigured firewall allowed excessive ICMP traffic into the network. • Internal network infrastructure and business operations were affected for approximately two hours.

	• Critical network resources became inaccessible to employees. • The organization recognized the need for stronger firewall configurations and regular security audits to identify vulnerabilities before exploitation.
Protect	• Configure firewall rules to rate-limit incoming ICMP traffic. • Enable source IP verification to reduce IP spoofing attacks. • Deploy an Intrusion Prevention System (IPS) to automatically block malicious ICMP traffic. • Establish regular firewall configuration reviews and security audits. • Provide security awareness training for IT staff on secure firewall management and network hardening.
Detect	• Deploy network monitoring software to identify unusual traffic spikes. • Use an Intrusion Detection System (IDS) to detect suspicious ICMP flood activity. • Enable centralized firewall logging and alerting for abnormal traffic patterns. • Utilize a SIEM solution to correlate logs and generate real-time security alerts.
Respond	• Immediately block malicious ICMP traffic at the firewall. • Isolate affected systems and temporarily disable non-critical services to reduce network load. • Notify management and IT personnel of the incident. • Analyze logs to identify the attack source and determine the scope of the incident. • Update the incident response plan based on lessons learned from the attack.
Recover	• Restore all network services after verifying the attack has ended. • Confirm firewall configurations and security controls are functioning

	correctly. • Monitor the network closely for any signs of recurring malicious activity. • Review and improve recovery procedures to reduce downtime during future incidents. • Document the incident and incorporate improvements into the organization's disaster recovery and business continuity plans.
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Reflections/Notes: This incident shows that a single misconfigured firewall can create a major security risk and that NIST helps with response